Shadow Node Theory v2.5.0 (v30):

Scale Invariance in the Node Satellization Algorithm — and a Universal Coupled Orbital Collapse Layer (ACO-A)

Empirical Verification Across 721 Real Cases, plus Collapse Evidence in Five Domains

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Pre-print v2.5.0 (framework v30) — not peer reviewed. Data and methodology available for review.

> ***Version note (v30).*** *This revision supersedes the previously posted SNT v2.3.1 (502-case corpus). A June 2026 audit found that the 502-case corpus contained ~188 synthetically generated b values (`np.random.normal()`) and an R² column with impossible values (down to –7.332); it has been* ***retired****. This version uses a **721-case corpus reconstructed entirely from verifiable primary sources** ($R^2 \in [0,1]$ for every case; reproducible from `reconstruction_real/`), and integrates a new* ***Coupled Orbital Collapse layer (ACO-A)****. The golden-ratio hypothesis (H-ϕ) was tested and* ***refuted across four rounds*** *(placebo control included) and is excluded from the main claims.*

Abstract

This paper presents Shadow Node Theory (SNT), a formal model of node satellization operating across three scales of systemic resolution — Micro (Atomic Node / individual), Meso (intra-national Fungal Network), and Macro (superorganism collision between nations and digital platforms). The central hypothesis holds that when two power nodes orbit in critical proximity, the node with greater accumulated advantage satellizes the historically dominant node through an algorithm whose dynamics follow a power law invariant to temporal scale and substrate: $R(t) = a \cdot t^b$, where b is the satellization velocity parameter.

SNT delivers four empirical contributions: (1) formalization of the Triple-Resolution Systemic Model with distinct applicability conditions across scales; (2) N-body matrix verification with Mexican INEGI data (32 federal entities, $b = –0.473$, $R^2 = 0.838$, $p < 0.001$), revealing that the binary model underestimated Tlaxcala's satellization gradient by 9.3×; (3) operationalization of the Atomic Sovereignty Index (ASI) on behavioral data from 4,774 users and 409,287 events (HackerEarth 2026, precision = 1.0, zero false positives); and (4) a corpus of **721 cases reconstructed from verifiable primary sources** spanning historical, economic, biological, astronomical and digital domains.

Three statistically robust findings emerge from the real corpus. First, **institutional friction is the dominant predictor of satellization velocity** (Spearman $\rho = -0.68$, $p = 2.5\times10$■■■, $n = 714$): the higher the friction, the lower the exponent. Second, **regime separation** — friction-free biological domains (b■ ≈ +0.95) satellize ~10× faster than friction-laden economic domains (b■ ≈ +0.09), Mann-Whitney $p = 2.4\times10$■■■. Third, **abrupt triggers produce faster satellization than gradual ones** (ratio 5.9×; Mann-Whitney $U = 24{,}802$, $p = 1.91\times10$■■, $n = 486$), stable across three successive corpus expansions (57 → 114 → 721 cases). Political sovereignty operates as a satellization brake mechanically equivalent to ecological mutual interdependence — both prevent definitive satellization by making complete node extinction destructive to the hub.

This revision adds a **Coupled Orbital Collapse layer (ACO-A)**: collapse is reformulated as an orthogonal axis (Δ), with a falsifiable hazard layer $h(\tau) > 0$ ("no system is eternal"), a three-factor taxonomy of collapse modes (friction × trigger × floor/ceiling), and a Principle of Least Friction unifying them, demonstrated with real data in five domains (finance, history, crypto, biology, astronomy). The model is accompanied by eight falsifiability criteria (RC1–RC8) plus collapse-axis criteria, a four-step diagnostic protocol, and a public replication package.

Keywords: complex systems, power law, satellization, scale invariance, preferential attachment, institutional friction, coupled orbital collapse, hazard function, leapfrog, Atomic Sovereignty Index, AI orchestration, digital ecosystems, regional inequality, Tlaxcala, Mexico.

JEL: O18, O33, D85, C63, O11, R11, C22.

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1. Introduction

1.1 The Scale Invariance Problem

Why do some regions remain poor despite decades of policy intervention? Why do some nations converge toward global leaders while others diverge irreversibly? These questions share a structural feature that aggregate models consistently fail to capture: the dynamics of resource extraction between proximate nodes operating at different hierarchical levels. Standard development economics models the poverty trap as a problem of insufficient capital accumulation. Shadow Node Theory proposes a different mechanism: satellization — the progressive extraction of productive residual energy from a peripheral node by a dominant hub — whose dynamics follow a power law invariant to temporal scale and substrate.

1.2 Theoretical Background

The preferential attachment mechanism (Barabási & Albert, 1999) establishes that scale-free networks emerge inevitably when new connections form with probability proportional to existing degree. SNT formalizes the directional flow of resources within such networks, quantifying the rate of divergence between hub and shadow node through the power-law exponent b. Leapfrogging theory (Brezis & Krugman, 1993) identifies conditions under which a peripheral node can bypass a dominant one through orthogonal-dimension investment. SNT extends this to three scales and formalizes the failure conditions the original model left undeveloped.

1.3 The Gap in the Literature

Three gaps motivate this work. First, no existing model quantifies satellization dynamics across historical cities, nation-states, intra-national regions and digital platforms within a unified formal framework. Second, the trigger taxonomy in leapfrogging theory collapses event types into a single category, missing the empirical distinction between abrupt and gradual triggers (with the real corpus confirming the stable hierarchy abrupt > gradual). Third, no operational index of cognitive sovereignty exists for the Atomic Node (individual) computable from observable behavioral data without self-report. A fourth gap, addressed by this revision, is the absence of a unified, falsifiable account of how systems **collapse** once the satellization relationship ends — the Coupled Orbital Collapse layer (Section 13).

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2. Formal Theoretical Framework

2.1 Definitions

A **Shadow Node** (or Peripheral Node) is any system component whose productive output is systematically extracted by a dominant hub over time, resulting in progressive divergence of productive capacity. A **Hub Node** is the dominant component that absorbs the residual productive energy of shadow nodes, increasing its own gravitational-mass advantage. **Critical Proximity** is the spatial, institutional, or digital proximity threshold below which the hub can extract resources at rates exceeding the shadow node's regeneration capacity.

The satellization ratio $R(t) = production\_hub(t) / production\_shadow(t)$ is the central observable. When $R(t)$ follows a power law $R(t) = a \cdot t^b$ with $b > 0$, satellization is active. When $b < 0$, convergence or leapfrog is occurring. The exponent b is the velocity parameter: $b > 0.45$ indicates accelerated satellization (abrupt-trigger class); $0.1 < b < 0.45$ indicates gradual satellization; $–0.1 < b < 0.1$ indicates approximate steady state; $b < –0.1$ indicates convergence or leapfrog.

2.2 Central Hypothesis

When two nodes orbit in critical proximity within the same system, the satellization ratio $R(t) = production\_hub(t) / production\_shadow(t)$ follows a power law $R(t) = a \cdot t^b$ where $b > 0$ represents the satellization velocity and t is elapsed time since the trigger event. This relationship is invariant to: (a) temporal scale — it holds from medieval city-pairs to digital platform rivalries; (b) production substrate — it holds for population, GDP per capita, behavioral event counts, and market share; and (c) system level — it holds for individuals, cities, regions, nations, and digital ecosystems.

2.3 Falsifiability Criteria

RC1 — Scalar Velocity: falsified if a technology is systematically adopted faster by institutions than by individuals (RC1a), or if a technology emerges without individual access that inverts the $TC\_micro < TC\_meso < TC\_macro$ hierarchy (RC1b). RC2 — Immune Response: falsified if hubs systematically adapt toward peripheral node capabilities rather than suppress them. RC3 — Qualitative Inextractability: falsified if a hub systematically neutralizes a node's knowledge differential through brain drain, reverse engineering, or deliberate saturation. RC4 — Dual

Minimum Threshold: falsified if a leapfrog sustains itself with either RQ or RL below the operational minimum. RC5 — Expansion Sequence: falsified if direct expropriation produces more stable outcomes than silent absorption for the same node class. RC6 — Irreversibility: falsified if a Shadow Node reverses satellization from inside the system without an exogenous trigger, with the hub operating normally.

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3. Methodology

3.1 General Design

The study follows a mixed quantitative design combining historical case analysis, N-body matrix modeling for the Mexican regional system, machine-learning validation on digital behavioral data, and survival analysis for the collapse layer. The unifying analytical pipeline consists of: (1) construction of the satellization ratio time series R(t) for each case pair; (2) log-log linearization; (3) ordinary least-squares regression to estimate parameters a and b; (4) Pearson correlation in log space for significance testing; and (5) interpretation via the SNT taxonomy.

3.2 Case Selection Criteria

Cases were included if they satisfy four conditions: (a) two nodes in critical proximity within the same system; (b) an identifiable trigger event or process; (c) production or population data at a minimum of four temporal points spanning at least 50 years (or 5 years for digital cases); and (d) the satellization process has either completed or is actively ongoing. No cases were excluded based on outcome direction — both satellization ($b > 0$) and convergence/leapfrog ($b < 0$) cases are included to avoid confirmation bias.

3.3 Data Sources (real corpus, v30)

The 721-case corpus is reconstructed entirely from verifiable primary sources: Maddison Project Database (country pairs and long historical series), INEGI and US Census (intra-national regions), Our World in Data / Johns Hopkins CSSE (COVID-19 spatial and parasite-host series), the Open Exoplanet Catalogue (planetary/stellar/multiplanetary), MacLulich (1937) and Elton & Nicholson (1942) (predator-prey), and the HackerEarth 2026 dataset (digital; N=4,774 users, 409,287 events, 141 event types). Every b is reproducible from public scripts.

Integrity: $R^2 \in [0,1]$ for all cases (zero negative, zero above 1); 89% are statistically significant ($p < 0.05$).

3.4 Analysis Pipeline

For each case pair, R(t) is computed at each temporal observation point. Time t is measured in years (or days for digital cases) elapsed since the trigger. The log-log transformation $\log(R) = \log(a) + b \cdot \log(t)$ enables OLS estimation. We report (a) the power-law exponent b; (b) R^2 of the log-log fit; (c) the Pearson correlation coefficient and p-value; and (d) the SNT classification. For the N-body Mexican matrix, we additionally compute the composite gradient — the total satellization force on each node accounting for all higher-level hubs.

3.5 Statistical Tests

Institutional-friction effect: Spearman correlation between an a-priori friction index (ordinal 0–3) and b per case (social/biological domains). Regime separation: Mann-Whitney U between friction-free biological and friction-laden economic domains. Trigger-type effect: Mann-Whitney U (abrupt vs gradual), excluding mutual-interdependence domains. Collapse layer: Spearman $\rho(b, \Delta)$ for orthogonality and Kaplan-Meier survival for the hazard. Significance threshold $\alpha = 0.05$; all tests in Python 3.11 (scipy.stats).

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4. Results — Historical Case Studies

4.1 Bruges → Antwerp (1300–1560)

Mechanism: physical infrastructure collapse (silting of the Zwin Canal, c.1490). R(t) = Antwerp/Bruges population follows a power law with b = +0.739 (R^2 = 0.868). Classified as accelerated satellization with abrupt trigger. Bruges had been the dominant commercial hub of Northern Europe for two centuries before the canal silting cut off maritime access; Antwerp, with open Scheldt access, absorbed the commercial network within two generations.

4.2 Toledo → Madrid (1528–1787)

Mechanism: pure political decree (Philip II's transfer of the imperial court, 1561). R(t) = Madrid/Toledo population: b = +0.694 (R^2 = 0.924), the best fit of the set. Toledo was the largest city in Castile; Madrid a village of fewer than

4,000 in 1528. Within 40 years of the court transfer, Madrid had surpassed Toledo — political decree alone, without geographic advantage, is sufficient to generate high-velocity satellization.

4.3 Portugal vs. Northwestern Europe (1535–1980)

Mechanism: Iberian Union (1580) combined with the accumulated Atlantic advantage of NW European powers. Trigger classified as hybrid. R(t) = NW Europe GDP pc / Portugal GDP pc: b = +0.060, R^2 = 0.123 — the low fit reflects an oscillatory process (Brazilian gold caused partial recoveries 1700–1750). The long-run trend is unambiguously divergent (the gap multiplied 3.5× between 1535 and 1913).

4.4 Tlaxcala → Puebla (1550–2022)

Mechanism: cumulative colonial extraction + differential industrialization. R(t) = Puebla/Tlaxcala GDP per capita: b = +0.184 (R^2 = 0.567), gradual satellization. Critically, the binary model underestimates Tlaxcala's actual satellization by measuring only the Tlaxcala–Puebla gradient; the N-body matrix (Section 8) reveals that 89.2% of extraction flows directly to Mexico City, making the true composite gradient 9.3× larger.

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5. Discussion: The Two-Speed Taxonomy

5.1 The Central Finding

The historical cases produce two statistically distinct classes of satellization dynamics. Abrupt-trigger cases (Bruges-Antwerp, Toledo-Madrid) generate exponents in the range b = 0.69–0.74. Gradual/hybrid cases (Portugal, Tlaxcala) generate exponents in the range b = 0.06–0.18. The ratio between classes is approximately 5.9×. In the full 721-case corpus this ordering is confirmed at scale (Mann-Whitney U = 24,802, p = 1.91×10■■, n = 486), stable across three successive corpus expansions. An earlier 57-case corpus had suggested hybrid triggers were fastest (p = 0.0098); the full corpus corrects that small-N artifact — the stable hierarchy is abrupt > hybrid > gradual.

5.2 Qualitative Interpretation

The key theoretical distinction is not the magnitude of the trigger but its reversibility. An abrupt trigger — infrastructure collapse, political decree — produces irreversible structural change the shadow node cannot compensate through internal resource mobilization. A gradual trigger — differential industrialization, slow technological diffusion — allows temporary adaptive responses that compress the b exponent without reversing the underlying dynamic.

5.3 Implications for Early Intervention

The b exponent has direct policy implications. A node with b = 0.70 faces a satellization horizon on the order of decades before the gap becomes structurally irreversible. A node with b = 0.18 has a longer window but the same terminal outcome absent intervention. The SNT diagnostic protocol (Section 11) provides a four-step procedure for estimating the current b, identifying the horizon of events, and designing orthogonal-dimension interventions that can generate b < 0 (convergence) without triggering the hub's immune response.

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6. Digital Validation: HackerEarth 2026

6.1 The Experiment

The HackerEarth 2026 dataset provides a behavioral event log for 4,774 users of the Zerve data-science platform across a 98-day window: 409,287 events, 141 event types. This constitutes a closed Meso-level system: the platform (hub) and its users (nodes), with resource flows measured as behavioral engagement metrics.

6.2 The Fractal Gap

The Composite Success Index v3 (CSI_V3), a weighted combination of tool diversity (0.40), platform lifetime (0.30) and Velocity of Diversification Rate (VDR, 0.30), reveals a fractal discontinuity. The Elite cohort (top 0.5%, n=24) shows a VDR 7,478× greater than the Basic cohort median (bottom 93.1%, n=4,444). This is not a power-law tail — it is a fractal discontinuity consistent with the SNT prediction of a Hub–Shadow Node separation following preferential attachment.

6.3 The 5-Event Wall

A Gradient Boosting Classifier on 284 event-type features achieves ROC-AUC = 0.9994 (test; 1.0000 in 5-fold CV). Users triggering fewer than five distinct event types have a >90% churn probability — the 5-Event Wall — detectable within the first session, the earliest possible intervention signal.

6.4 SHAP Rankings and the Cognitive Leapfrog

The top SHAP-ranked predictor is AI-agent orchestration (agent_accept_suggestion, SHAP proxy ~0.5). Users who delegate execution to the AI agent rather than executing linearly are the strongest predictor of Elite-trajectory behavior — the cognitive leapfrog: transitioning from linear execution to agent orchestration, a dimension where accumulated hub advantage does not apply and which is currently accessible to new entrants.

6.5 Extension to the Enterprise Domain

The HackerEarth case is the first empirical demonstration of SNT applied to a closed enterprise ecosystem. The applicability condition is not sector or size but the availability of structured behavioral event data measuring inter-node resource flows. Five metrics of the Composite Enterprise Sovereignty Index (CSIE) — event volume per node, sustained activity, response time, functional diversity, and failure resilience — identify satellization within organizational structures before it becomes structurally irreversible.

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7. SNT Triple-Resolution Systemic Model

The model extends the original binary model across three scales with distinct dynamics, actors, and incompatible competitive rules. The five-level taxonomy and INEGI 2022 verification confirm that the Mexican national system operates under preferential attachment; trajectory vectorization for eight entities (1940–2022) documents the first cases of successful leapfrog within the system: Querétaro ($b = -0.155$, $p < 0.01$) and Nuevo León ($b = -0.058$, $p < 0.001$).

7.1 Micro Resolution — The Atomic System. The base scale of processing and survival. Resources divide into Quantitative (RQ, extractable: capital, time, infrastructure) and Qualitative (RL, inherent: knowledge, skills, cognitive maturity). RL cannot be directly extracted but degrades through disuse when RQ scarcity prevents its maintenance. Leapfrog requires two parallel dimensions: Intrapersonal (DI, mandatory base) and Professional (DP, visible leap).

7.2 Meso Resolution — The Intra-national Fungal Network. A closed ecosystem delimited by geopolitical or institutional jurisdiction. The Central Hub administers the network through continuous extraction of residual energy from Shadow Nodes. The hub is practically immovable from inside; its immune response activates based on growth direction, not size. The hub expands via silent absorption, peaceful agreement, or expropriation — in that order of energetic-cost preference.

7.3 Macro Resolution — The Superorganism Collision. Competition between complete fungal networks; no central hub arbitrates. Relative position is set by Gravitational Mass (MG: total GDP, population density, technological level, territorial area). The real brake on aggressive expansion is the internal node network, not international regulators. The Atomic Node never fully escapes the Macro system: its legal and fiscal existence is anchored to its resident superorganism.

7.4 Cross-Scale Interaction Principles. Cascade Transmission: Macro events impact all three systems in descending cascade (Macro → Meso → Micro), at a speed depending on each node's dimensional independence. Scalar Velocity: TC_{micro} (hours–months) < TC_{meso} (months–years) < TC_{macro} (decades–generations), differing by 10–100× per level. Speed is the structural advantage of the Atomic Node.

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8. Empirical Verification: N-Body Matrix — Mexico

The five-level taxonomy is verified with INEGI 2022–2023 data for the 32 federal entities. Level 0 (Mexico City): 14.8% of national GDP. Level 1 (9 secondary attractors): 41.0%. Level 2 (8 logistic-bypass nodes): 20.2%. Level 3 (11 shadow nodes): 16.8% with the largest number of entities. Level E (3 exogenous anomalies): 4.3%. The power-law fit confirms preferential attachment: $b = –0.473$, $R^2 = 0.838$, $p < 0.001$.

The composite gradient of Tlaxcala is the central N-body result: the binary model measured w_{ij}(Tlaxcala→Puebla) = 26.2k MXN; the N-body matrix reveals w_{ij}(Tlaxcala→Mexico City, long-range) = 216.8k MXN. Total composite gradient: 243.0k MXN — the binary model underestimated Tlaxcala's satellization by 9.3×; 89.2% of extraction flows directly to Mexico City, bypassing the intermediary.

Trajectory vectorization for eight entities (1940–2022) reveals two natural groups: Satellization ($b > 0$): Chiapas (+0.229), Oaxaca (+0.176), Guerrero (+0.176), Veracruz (+0.181), Tlaxcala (+0.147), Puebla (+0.116); Convergence ($b < 0$): Querétaro (–0.155, R^2=0.782) and Nuevo León (–0.058, R^2=0.935) — the first documented leapfrog cases within the national system (Querétaro via aerospace manufacturing, Nuevo León via independent export manufacturing).

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9. Limitations

Original corpus limitations remain: data uncertainty (±20%) for pre-1820 historical estimates, low R^2 for the Portugal case (oscillatory process), possible reverse causality in the digital case, selection bias in case choice, and sensitivity to trigger definition in gradual cases. The Triple-Resolution Model's Micro and Macro modules are conceptual frameworks with partial operationalization; variables RQ, RL, DI, DP, and MG have proposed measurement criteria not yet validated with structured data series. The ASI is operationalized with precision = 1.0 on HackerEarth 2026 but requires replication on independent datasets. The Ck Coherence Factor has a verified neurological mechanism (Friston 2010) but its operationalization is untested.

On the corpus: this version retires the previously posted 502-case corpus, which contained synthetic values, and replaces it with 721 cases reconstructed from verifiable primary sources ($R^2 \in [0,1]$; 89% significant). The collapse layer (Section 13) is correlational and, on the crypto side, based on small n; it is framed as a strong hypothesis, not causal proof.

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10. Falsifiability Criteria

RC1 — Scalar Velocity: falsified if a technology is systematically adopted faster by firms than by individuals (RC1a), or if a technology emerges without individual access that inverts the TC hierarchy (RC1b). RC2 — Immune Response: falsified if hub adaptation toward the peripheral node is documented more frequently than suppression. RC3 — Qualitative Inextractability: falsified if hubs systematically neutralize node knowledge differentials. RC4 — Dual Minimum Threshold: falsified if a leapfrog sustains itself with either dimension below minimum. RC5 — Expansion Sequence: falsified if direct expropriation produces more stable states than silent absorption. RC6 — Irreversibility: falsified if a Shadow Node reverses satellization endogenously without an exogenous trigger. RC7 — ASI Operationalization: falsified if users with ASI > 1 show performance comparable to users with ASI < 0.5 in tasks requiring cognitive sovereignty (current validation precision = 1.0; requires replication). RC8 — Mutual-interdependence brake: falsified if predator-prey or sovereign-state systems produce sustained $b > 0.5$ without exogenous perturbation.

Collapse-axis criteria (new). RC-Δ1 — Orthogonality: falsified if corr(b, Δ) is significantly different from zero across paired cases (first test: crypto n=11, $\rho = +0.009$, not refuted). RC-Δ2 — Friction governs collapse shape: falsified if resolution friction does not predict Δ (first test: 2008 cohort n=6, $\rho = -1.000$). RC-Δ3 — Hazard positivity: falsified if a system with hazard = 0 is found.

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11. Diagnostic Protocol

SNT is prescriptive as well as descriptive. Four steps to apply the model to any real system. **Step 1 — Level Classification:** collect node production data; verify whether the distribution follows a power law (log-log fit, $R^2 > 0.7$, $p < 0.05$); classify in the five-level taxonomy. **Step 2 — Gradient Calculation:** calculate w_{ij} for each hub extracting from the node; compute the composite gradient if there are multiple hubs (the Tlaxcala case shows the long-range gradient can be 8.3× larger than the direct one). **Step 3 — Event Horizon Estimation:** fit the historical trajectory; if $b > 0$, estimate $t_{horizon}$; if $b < 0$, identify the mechanism and verify its sustainability. **Step 4 — Orthogonal Dimension Identification:** search for dimensions where the hub has not invested in 5–10 years, where the node has a measurable initial advantage, and which have preferential-attachment potential; verify they do not require hub-controlled infrastructure; then design the intervention (address the critical deficiency at minimum, build capacity without triggering the immune response, execute the leapfrog when the window is open).

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12. Corpus of 721 Real Cases — Definitive Findings

The empirical corpus comprises 721 cases reconstructed entirely from verifiable primary sources, replacing the previous 502-case corpus (retired after an audit found ~188 synthetic b values and an impossible R^2 column). Distribution by domain (friction, n, b■): A Cities (medium, 4); B Country pairs Maddison (high, 446, +0.092); C Regions INEGI+Census (high, 24, +0.091); D Digital (low, 3, –1.364); E1 Biological invasion (none, 4, +2.891); E2 Predator-prey (high, 2, +0.145); E3 Parasite-host COVID (none, 234, +0.912); F1–F3 Astronomical (medium/low, 4). Total 721, global b■ = +0.366; integrity $R^2 \in [0,1]$ for all; 89% significant.

Finding 1 — Institutional friction predicts satellization (central result).

Spearman correlation between the a-priori friction index and b per case (social/biological domains, n = 714): ρ = –0.68, p = 2.5×10■■■. The higher the institutional friction, the lower the satellization exponent.

Finding 2 — Regime separation. Friction-free biological domains (E1+E3) produce b■ ≈ +0.95; friction-laden economic domains (A+B+C) produce b■ ≈ +0.09. Mann-Whitney U = 103,538, p = 2.4×10■■■. Systems with no institutional brake satellize ~10× faster. This is the central empirical result, and it is stronger with real data than with the previous synthetic values.

Finding 3 — Abrupt triggers are faster than gradual. Ratio 5.9×, Mann-Whitney U = 24,802, p = 1.91×10■■ (n = 486, excluding mutual-interdependence domains), stable across three successive corpus expansions (57 → 114 → 721). The earlier 57-case "hybrid fastest" result (p = 0.0098) is a small-N artifact corrected by the full corpus; the stable hierarchy is abrupt > hybrid > gradual.

Finding 4 — Political sovereignty is as effective a brake as ecological interdependence. Sovereign country pairs (b■ ≈ +0.09) and predator-prey systems (E2, b■ = +0.145) are statistically indistinguishable — two distinct mechanisms anchoring b near zero because complete node extinction would destroy the hub. A herring shoal cannot negotiate with a school of mackerel; a Hot Jupiter does not apply for regulatory approval before clearing the inner solar system. When resource transfer is direct and unmediated, b exceeds 1 regardless of substrate.

Finding 5 — Modeling regimes. The power law is the best description where friction is low (epidemics E3, biological invasion E1); under high friction (countries) exponential and linear models compete. The exponent b is a descriptive, cross-domain comparable metric — not a claim that the power law is the only generative model in all domains.

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13. Coupled Orbital Collapse Layer (ACO-A)

The Orbital Collapse Architecture ceases to be a separate module and is reformulated as a **universal, transversal layer** of SNT: collapse is an **orthogonal axis** that can activate in any system, in any domain, at any point of its trajectory. A single principle (least friction) generates distinct collapse modes depending on boundary conditions, demonstrated with real data in five domains.

13.1 Two orthogonal axes (b ⊥ Δ). Each system is a pair of independent coordinates. Axis 1 — Satellization: $R(t) = a \cdot t^b$, how dominance evolves while the coupled relationship runs. Axis 2 — Collapse: $A(\tau) = c \cdot \tau^\Delta$, with τ = time since functional extinction; Δ measures the speed/shape of absorption once the hub collapses. Collapse does not wait for the satellization cycle to end (different clock, $\tau \neq t$). Falsifiable prediction: $corr(b, \Delta) \approx 0$. First test (paired crypto, $n = 11$): Spearman $\rho(b_{rise}, \Delta_{fall}) = +0.009$ ($p = 0.98$) — consistent with orthogonality.

13.2 Hazard layer h(τ). "No system is eternal" = $h(\tau) > 0$ for every system (refutable if a system with hazard = 0 is found). First estimate (crypto cohort, $n = 41$; functional extinction = price < 1% of all-time high): 15 extinctions across the whole age range (0.27–8.6 years), no death-free era; Kaplan-Meier declining; hazard positive and rising with age — consistent with $h(\tau) > 0$. Caveats: survivorship bias (true hazard higher), age/calendar confound, limited n.

13.3 Taxonomy of collapse modes (three factors). Governed by friction × trigger × (floor/ceiling on magnitude): Regulated Orbital Decay (high friction → smooth power law or exponential, non-accelerating; 2008 cohort R^2=0.85–0.99, Rome/USSR, solar flare R^2=0.975, TDE R^2=0.84); Cracquelure Decay (friction≈0 + gradual → erratic fragmentation; EOS R^2=0.10–0.70); Floor-Arrested (friction≈0 + abrupt + floor → power law to a residual floor; FTX R^2=0.875); Catastrophic Cliff (friction≈0 + abrupt + no floor → super-exponential; LUNA, 5.6 orders of magnitude in 11 days); Logistic Sweep (bounded magnitude → S-curve; Delta→Omicron k=0.22/d).

13.4 Principle of Least Friction (unifier). Every collapse follows the trajectory that minimizes integrated friction (variational family: Fermat, least action, minimum dissipation) — a gradient flow over a stability landscape. Falsifiable version: the realized collapse has lower integrated friction than counterfactual trajectories (WaMu via the pre-arranged FDIC channel = least friction → 21 h; Lehman without it → slow fragmentation, 30,681 h; range ~1,460×, monotonic with the degree of regulatory intervention).

13.5 Results with real data (four roadmap items). (1) Friction operationalized: within the 2008 financial cohort (n=6), resolution-channel friction (ordinal 1–6) vs Δ: Spearman $\rho = -1.000$, $p < 0.001$ — more friction, more frontal and orderly absorption. (2) Orthogonality $b \perp \Delta$: crypto n=11, $\rho = +0.009$ (§13.1). (3) Unbounded biology: the Omicron wave in absolute counts (South Africa, JHU) decays smoothly exponential ($R^2 = 0.96$, e-fold ~22 d), NOT a cliff — epidemiological feedback is intrinsic friction. (4) Hazard $h(\tau) > 0$ (§13.2). Connection to the central finding: institutional friction predicts b ($\rho = -0.68$) and also governs the shape of Δ — it is the lever for both axes.

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14. Dialogue with the Literature

Barabási & Albert (1999): SNT extends preferential attachment by quantifying satellization velocity through the exponent b and proposing a five-level functional taxonomy. INEGI verification confirms the predicted distribution ($b = -0.473$, $R^2 = 0.838$, $p < 0.001$). **Watts & Strogatz (1998):** SNT adds the directional dimension of resource flow to the clustering coefficient — two nodes may be close in connection distance but at radically different hierarchical levels. **Holland (1995):** SNT specifies satellization as a recurring emergent dynamic with a mathematically predictable trajectory within Complex Adaptive Systems. **Friston (2010):** SNT extends the Free Energy Principle beyond the individual brain to social systems; the hub immune response and the Coherence Factor Ck are manifestations of the same principle at different scales. **Brezis & Krugman (1993):** SNT extends technological leapfrogging to three scales and formalizes the failure conditions the original model did not develop. **Catastrophe and resilience theory (Thom 1972; Waddington 1957; Holling 1973; Lenton et al. 2008):** the collapse layer's stability-landscape language connects SNT to fold catastrophes, the epigenetic landscape, ecological "ball-in-cup" resilience, and climate tipping points.

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15. Conclusions

15.1 What SNT demonstrates. The satellization cycle — from dependent child node to peer to hub of new children — operates across all domains. The five-level taxonomy is verifiable with INEGI data and follows a power law; the binary model underestimated Tlaxcala's satellization by 9.3×; the first documented leapfrog cases within the Mexican national system are verified (Querétaro $b = –0.155$, Nuevo León $b = –0.058$); the ASI is operationalized with precision = 1.0 on 4,774 users. With the real 721-case corpus, **institutional friction is the dominant predictor of b **(ρ = –0.68, p = 2.5×10■■■) and** also governs the shape of collapse (Δ)** — a single variable unites how a system dominates and how it collapses. The golden-ratio hypothesis (H-ϕ) was tested and refuted across four rounds and is excluded from the claims.

15.2 What SNT does not demonstrate. That leapfrog is always possible for any node. The model formalizes viability conditions and failure mechanisms but does not guarantee success. The Micro and Macro modules require independent empirical validation; the ASI requires replication beyond HackerEarth 2026; the collapse layer is correlational and requires larger cohorts and cross-domain tests.

15.3 Future research lines. Validation of the Micro Module with longitudinal individual trajectory data; ASI operationalization with other platforms; extension of the N-body matrix to other national systems; cross-domain orthogonality test for $b \perp \Delta$; larger survivorship-unbiased cohorts for $h(\tau)$; pre-registration before claiming causality.

15.4 The major implication. Not theoretical but practical. If satellization follows a predictable algorithm with an identifiable failure taxonomy, it is intervene-able. For Tlaxcala: 89.2% of the gradient comes not from Puebla but from Mexico City — any strategy targeting only the Tlaxcala–Puebla relationship solves 10.8% of the problem. For the Atomic Node: the cognitive leapfrog — orchestrating AI agents rather than executing tasks linearly — is the first dimension in recent history where the accumulated advantage of dominant nodes does not apply directly, and the HackerEarth 2026 evidence suggests that window is open now.

"The satellization algorithm is predictable. The leapfrog failure taxonomy is known. What follows is a decision that no model can make for the node."

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Collapse-layer data sources: Yahoo Finance (LUNA, FTT, EOS); NOAA SWPC GOES (solar X-ray); NASA IRSA / ZTF (TDE AT2019qiz); CoV-Spectrum / LAPIS (SARS-CoV-2 variants); SEC, FDIC, Federal Reserve, SIGTARP (2008 cohort).
